

PRODUCT LITERATURE

1. Product Name:

PVP-IVERMECTIN 0.5% W/V POUR-ON

2. Each 1000 ml pour-on contains:

Active Ingredient:

Ivermectin BP	5.00 g
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Preservatives:

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| 1. Propyl Paraben BP | 200.00 mg |
| 2. Methyl Paraben BP | 2.00 g |

3. Product Description:

Clear, colourless to pale yellow solution.

4. Target Species:

Cattle.

5. Pharmacodynamics

Ivermectin is a broad-spectrum endectocide of the avermectin family. Ivermectin is isolated after purification and hydrogenation of the avermectin-family compounds which are obtained from the fermentation of the soil organism, *Streptomyces avermitilis*.

Ivermectin is a macrocyclic-lactone derivative which has a broad and potent antiparasitic activity against nematodes and arthropods. It acts by inhibiting nerve impulses. Ivermectin binds selectively and with high affinity to glutamate-gated chloride-ion channels which occur in invertebrate nerve and muscle cells. This leads to an increase in the permeability of the cell membrane to chloride ions with hyperpolarization of the nerve or muscle cell, resulting in paralysis and death of the relevant parasites. Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA). The margin of safety for compounds of this class is attributable to the fact that mammals do not have glutamate-gated chloride channels. The macrocyclic lactones

have a low affinity for other mammalian ligand-gated chloride channels and they do not readily cross the blood-brain barrier.

6. Pharmacokinetics:

After administration of the recommended dosage to cattle, varying inter-individual ivermectin plasma levels were observed with mean values of C_{max} and t_{max} of 17 ng/ml and 170 h, respectively.

After topical administration of 0.5 mg of ivermectin per kg of bodyweight, liver and fat (the target tissues) generally had the highest residues. Excretion occurs mainly through faeces and, in lesser proportion, via urine.

7. Environmental Properties:

EXTREMELY DANGEROUS TO FISH AND AQUATIC LIFE. Do not contaminate surface waters or ditches with the product or used container.

8. Indication:

Treatment of gastro-intestinal nematodes, lungworms, warbles, chorioptic and sarcoptic mange mites, sucking and biting lice of beef and non-lactating dairy cattle.

Gastro-intestinal roundworms *Ostertagia ostertagi* (L4, adults and inhibited stages), *Haemonchus placei* (L4, adults), *Trichostrongylus axei* (L4, adults), *Trichostrongylus colubriformis* (L4, adults), *Cooperia* spp. (adults), *Cooperia punctata* (adults), *Cooperia oncophora* (adults), *Oesophagostomum radiatum* (L4, adults), *Strongyloides papillosus* (adults), *Trichuris* spp. (adults) Lungworms (adults and 4th-stage larvae), *Dictyocaulus viviparus*, Warbles (parasitic stages), *Hypoderma bovis*, *Hypoderma lineatum*, Mites, *Sarcoptes scabiei* var. *bovis*, *Chorioptes bovis*, Lice- Sucking lice, *Linognathus vituli*, *Haematopinus eurysternus*- Biting lice, *Damalinea bovis*

The product, at the recommended use level of 500 µg/kg of ivermectin, has a persistent activity on: *Dictyocaulus viviparus*: for up to 28 days, *Ostertagia* spp.: for up to 21 days, *Oesophagostomum radiatum*: for up to 21 days, *Cooperia* spp.: for up to 14 days, *Trichostrongylus axei*: for up to 14 days.

The product helps in the control of the mange mite *Chorioptes bovis* but complete elimination may not occur.

The product has also a persistent activity on the horn fly (*Haematobia irritans*) for 28 days; partial efficacy may last for up to 35 days post application. Occasionally, variable activity may be observed against *Haemonchus placei* (L4), *Cooperia* spp., *Trichostrongylus axei* and *Trichostrongylus colubriformis*.

9. Recommended Dose:

1 ml per 10 kg of bodyweight (based on a recommended dosage level of 500 µg/kg of ivermectin).

This product is for pour-on used i.e. it is topically applied.

This product is equipped with a separate graduated applicator can. Care should be taken when transferring the product into the graduated applicator can or other suitable dosing device. Place the graduated applicator can in a well-ventilated area or outdoors, on a flat surface when transferring the product. A clean plastic funnel is preferably used to ease the transfer and avoid spillage. Fill the required amount of the product into the graduated applicator can.

Tilt and slowly squeeze the graduated applicator can to apply the product. The product should be applied along the mid-line of the back in a narrow strip between the withers and tailhead.

To ensure correct administration of a correct dose:

- i. bodyweight should be determined as accurately as possible;
- ii. accuracy of the dosing device should be checked.

Important: Keep the product upright during storage. Close container when not in use and store in an upright position.

10. Mode of Administration:

Topical

11. Contraindications:

Do not use in animals known to be hypersensitive to the active substance or to any of the ingredients. Do not inject or give orally. This product is for application to skin surface only. Do not use in dairy cows during lactation or the dry period and in beef cows during the lactation period when milk is intended for human consumption. Do not use in pregnant dairy heifers within 60 days prior to calving.

12. Warning & Precautions:

Assess bodyweight as accurately as possible before calculating the dosage. Cattle should not be treated when the hair or hide is wet. Rain falling on animals less than two hours after dosing may result in reduced efficacy. However, under such conditions, efficacy of the product against infections of *Ostertagia ostertagi* or *Dictyocaulus viviparus* in cattle may be maintained. The influence of extreme weather conditions on the long-term performance (persistent activity) of the product is not known. Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

Too frequent and repeated use of anthelmintics from the same class, over an extended period of time. Underdosing, which may be due to underestimation of body weight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used. Therefore the use of this product should be based on local (regional and farm) epidemiological information about susceptibility of the gastrointestinal nematodes and recommendations on how to limit further selection for resistance to anthelmintics.

Special precautions for use in animals

Do not apply to areas of skin which have mange scabs or other lesions or to areas contaminated with mud or manure.

Avermectins may not be well tolerated in all non-target species. Cases of intolerance with fatal outcome are reported in dogs, especially collies, old English sheepdogs and related breeds or crosses and also in turtles/tortoises.

The product is effective in all hypodermosis stages. However, it is important to treat on time (at the end of warble-fly season). The elimination of *Hypoderma* larvae may cause negative reactions on the host, when they are found in vital areas. Killing *Hypoderma lineatum*, if found in perioesophageal tissue, may cause salivation and tympanism. Killing *Hypoderma bovis*, if found in the vertebral canal, may cause unsteadiness or paralysis. Cattle should be treated before or after those stages of warble flies.

Special precautions to be taken by the person transferring the veterinary medicinal product from the immediate container into the graduated applicator can and administering the veterinary medicinal product to animals

HIGHLY FLAMMABLE - keep away from heat, sparks, open flame or other sources of ignition. Care should be taken when transferring the product into the graduated applicator can or other suitable dosing device. The product may be irritating to human skin and eyes and the user should be careful not to apply it to himself or other persons. Operators should wear nitrile rubber gloves, rubber boots and a waterproof coat when transferring and applying the product. Dosing and transferring devices and protective clothing should be washed after use. If accidental skin contact occurs, wash the affected area immediately with soap and water. If accidental eye exposure occurs, flush the eyes immediately with water and get medical attention. Do not smoke, drink or eat while handling the product. Wash hands after use. Use only in well-ventilated areas or outdoors.

Other precautions

Treated cattle should not have direct access to ponds, streams or ditches for 14 days after treatment.

13. Drug Interactions:

Do not combine treatment with vaccination against lungworm infection. If vaccination is intended, an interval of at least 28 days before or after the date of vaccination should be taken into account.

14. Pregnancy & Lactation:

Do not use in dairy cows during lactation or the dry period and in beef cows during the lactation period when milk is intended for human consumption. Do not use in pregnant dairy heifers within 60 days prior to calving

15. Side effects:

None known

16. Symptoms & Treatment of Overdose:

Studies have demonstrated a wide safety margin. No sign of toxicity appeared in trials up to 5 mg/kg (10 times the recommended dose rate). No antidote has been identified.

In case of overdose, symptomatic treatment should be given. The symptoms of overdose can be trembling, convulsions and coma.

17. Withdrawal Period:

Cattle

Meat and offal: 28 days.

Not permitted for use in lactating cows producing milk for human consumption. Do not use in pregnant cows, which are intended to produce milk for human consumption, within 60 days of expected parturition.

18. Storage Conditions:

Store below 30° C in dry place, protected from light.

19. Dosage Form:

Liquid (Topically Applied)

20. Packaging:

Packed in 1 L HDPE plastic bottle container with caps. Equipped with a separate 150 ml PVC plastic bottle graduated applicator can along with pointed cap.

21. Disposal of Container:

To be disposed as per local authority requirement.

22. Expiry Date:

2 years from date of manufacturing.

Manufactured & Distributed by :

Pelangi Vet Pharma Sdn Bhd. (665097K)

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